

ALGORITHM – VEHICLE RENTAL MANAGEMENT SYSTEM (WEEK 1)

Purpose: To manage vehicle records using basic operations like Add, View, Update and Delete using ArrayList and Encapsulation.

1 Start

- Start the program.

2 Create ArrayList

- Create an ArrayList to store Vehicle objects.

3 Display Menu

- Display the menu with the following options:

1. Add Vehicle
2. View Vehicles
3. Update Vehicle
4. Delete Vehicle
5. Exit

4 Read Choice

- Read the user's choice.

5 Add Vehicle

- If choice is 1:
 - Read Vehicle ID.
 - Read Vehicle Name.
 - Read Rent Per Day.
 - Create a Vehicle object.
 - Add the object to the ArrayList.

6 View Vehicles

- If choice is 2:
 - If ArrayList is empty, display “No Vehicles Available”.
 - Otherwise, display all vehicle details.

7 Update Vehicle

- If choice is 3:
 - Read Vehicle ID to update.
 - Search the vehicle in the ArrayList.
 - If found, read new Vehicle Name and Rent Per Day.
 - Update the vehicle using setter methods.
 - Display “Vehicle Updated Successfully”.
 - Otherwise, display “Vehicle Not Found”.

8 Delete Vehicle

- If choice is 4:
 - Read Vehicle ID to delete.
 - Search the vehicle in the ArrayList.
 - If found, remove the vehicle from the ArrayList.
 - Display “Vehicle Deleted Successfully”.
 - Otherwise, display “Vehicle Not Found”.

9 Exit

- If choice is 5, terminate the program.

10 Repeat

- Repeat steps 3 to 9 until the user chooses to exit.

11 End

- Stop the program.

